Lecture 12: Introduction to Instrumental Variables

So far, we have focused on ordinary least squares. Least squares allows us to recover causal effects, but only under an orthogonality condition: we need the regressors to be uncorrelated with the error term for OLS to recover true causal effects. Unfortunately, in practice this is not always such a reasonable assumption: we are often interested in things that are going to naturally be correlated with the error term.

Today we will introduce an alternative approach, called instrumental variables (IV). The key idea behind instrumental variables is that if we are interested in the effect of X on Y, then we need to find a third variable, Z, that we can use to generate "clean" variation in X. We call this third variable, Z, our instrument. Whereas in OLS we needed an orthogonality condition for X, in IV we will rely on an orthogonality condition for Z.

Setup

Suppose that we are interested in the following regression:

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$

where Y_i is log earnings, and X_i is years of schooling. We have shown that OLS will only recover the true causal effect if the regressor is uncorrelated with the error term: $\text{Cov}(X_i, \varepsilon_i) = 0$. Unfortunately, we might often think that is not the case: in this case, for example, we might think that people complete more years of schooling if they are smart and/or hardworking, and smart, hardworking people tend to do better in the labor market, holding years of education fixed. In this case, we would say that our regressor, X_i, is **endogenous**: it is itself caused by other things that affect the outcome, and thus an OLS regression of Y_i on X_i will not recover the true causal effects.

What if instead, we had a variable, Z_i, which we thought was correlated with X_i, $\text{Cov}(X_i, Z_i) \neq 0$, and which was uncorrelated with the error term, $\text{Cov}(Z_i, \varepsilon_i) = 0$. For example, Z_i might be a randomly assigned college scholarship, or an admissions offer that was given by lottery. We could then use this to recover the causal effect, β:

$$\begin{aligned}
\text{Cov}(Y_i, Z_i) &= \text{Cov}(\alpha + \beta \cdot X_i + \varepsilon_i, Z_i) \\
&= \beta \cdot \text{Cov}(X_i, Z_i) + \text{Cov}(\varepsilon_i, Z_i) \\
&= \beta \cdot \text{Cov}(X_i, Z_i) \\
\implies \beta &= \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(X_i, Z_i)}
\end{aligned}$$

If we have a valid instrument, we can use it to recover the causal effect of X_i, even if X_i is correlated with the error!

Notice also that we can express this as the ratio of two regression coefficients. Suppose that we ran the following regressions, which we will refer to as the **reduced form** and **first stage** regressions.

$$\begin{aligned}
\text{Reduced Form: } Y_i &= \kappa + \rho \cdot Z_i + \omega_i \\
\text{First Stage: } X_i &= \gamma + \pi \cdot Z_i + u_i
\end{aligned}$$

We have:

$$\begin{aligned}
\beta &= \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(X_i, Z_i)} \\
&= \frac{\text{Cov}(Y_i, Z_i) / \text{Var}(Z_i)}{\text{Cov}(X_i, Z_i) / \text{Var}(Z_i)} \\
&= \frac{\rho}{\pi} = \frac{\text{Reduced Form}}{\text{First Stage}}
\end{aligned}$$

which shows that, in the single-instrument-single-regressor setting, the IV coefficient is the ratio of the regression of Y on Z (reduced form) to the regression of X on Z (first stage). This representation will be very useful to us as we are building our understanding of IV.

Conditions for Consistency and Asymptotic Normality of IV

We will now discuss the conditions that will guarantee consistency of IV. These same conditions will also give us asymptotic normality of IV. Below we provide the conditions for the single-regressor, single-instrument case, and we will discuss the general case next time.

Assumption (IID Sampling)**.** *Each observation is drawn independently and from the identical distribution as the other observations.*

Assumption (Relevance)**.** *The instrument covaries with the regressor, Cov* $(X_i, Z_i) \neq 0$*.*

Assumption (Orthogonality Condition)**.** *The error term,* ε_i*, has zero mean* $\mathbb{E}[\varepsilon_i] = 0$*, and is uncorrelated with the instrument, Cov* $(Z_i, \varepsilon_i) = 0$*.*

The IID sampling condition serves the same role as in OLS: it ensures that we can use the law of large numbers to show that sample covariances converge to the population covariances, and similarly it allows us to use the central limit theorem to prove asymptotic normality. Instead of no perfect collinearity, we now have the relevance condition: the instrument must be correlated with the regressor. This ensures that the instrument creates variation in the regressor, which we can use to estimate β. We will discuss the multiple regressors case in our next lecture, but if there are multiple regressors then there will be multiple instruments, and relevance will require that those instruments must create distinct variation in each of the regressors, much like our earlier requirement of no perfect collinearity for OLS. Finally, our new orthogonality condition states that, rather than the regressor being uncorrelated with the error term, we instead need the instrument to be uncorrelated with the error term.

Understanding Orthogonality

Although we state the orthogonality condition simply as the uncorrelatedness of the instrument and the error, $\text{Cov}(Z_i, \varepsilon_i) = 0$, it is really best understood as two separate concepts. The first is **as-good-as-random assignment**. This says that the instrument is assigned in a way such that it is uncorrelated with the error term. We can get this either from actual random assignment (e.g. in a randomized controlled trial, like the de Mel et al. example), or from something that is not explicitly randomized, but which we think might still be uncorrelated with the error term.

The second component of the orthogonality condition is the **exclusion restriction**. To satisfy the exclusion restriction, it must be the case that the instrument only affects the outcome through its effect on the regressor: it must have no direct effect on the outcome. The term "exclusion restriction" comes from the idea that we are writing $Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$, and are excluding Z_i from that model. It is possible for an instrument to be randomly assigned but not be excludable. For example, suppose that our instrument for college attendance was a randomly assigned college scholarship, which came bundled with other benefits such as summer internships and job placement assistance. The scholarship program would likely fail the exclusion restriction: the program would affect earnings not just through its effect on years of schooling, but also directly through the internship and job placement assistance.

Testable vs. Non-Testable Assumptions

Relevance is an important assumption, but we don't spend much time worrying about it. The reason is that this result is testable: we can observe both X_i and Z_i, and so it is easy to test the null hypothesis $\pi = 0$, which would indicate that the instrument is irrelevant. If we can reject that hypothesis at a low significance level, then we can be confident that our instrument is in fact relevant.

Orthogonality, however, is not really testable. We do not observe ε_i, and so we cannot test whether it has zero covariance with Z_i. Note that we cannot just replace ε_i with the $\hat{\varepsilon}_i$ that comes out of the IV

regression: we obtained these residuals by assuming they were uncorrelated with the instrument, so IV will simply select the β that makes the error term and the instrument uncorrelated. We can do things like including variables as controls or testing if the instrument is correlated with other regressors that we think are confounds, but it is not possible to fundamentally test the orthogonality condition. Our assumption of orthogonality must thus come from our understanding of economics and the world.

Consistency of IV

Under our three assumptions for instrumental variables, we can show that IV is consistent:

Theorem (Instrumental Variables is Consistent)**.** *Suppose that each observation is sampled iid, the instrument is relevant, and the orthogonality condition holds ($\mathbb{E}[\varepsilon_i] = 0$ and $Cov(Z_i, \varepsilon_i) = 0$). Then, IV is consistent: $\hat{\beta}^{IV} \xrightarrow{p} \beta$, where $\hat{\beta}^{IV} = \frac{\widehat{Cov}(Y_i, Z_i)}{\widehat{Cov}(X_i, Z_i)}$.*

The proof is fairly straightforward: IID sampling guarantees that the covariances converge to the true population covariances, relevance guarantees that the denominator is not zero, and then orthogonality ensures that the ratio gives us the true β, by the argument we already gave in the setup.

Examples

Now that we have laid out the theory, we will go over a few different examples of IV in practice.

Returns to Capital (de Mel, McKenzie, and Woodruff 2008)

We previously discussed the de Mel, McKenzie, and Woodruff (2008) setting. After showing in earlier specifications that they could not reject that "a rupee is a rupee," they combined the grants into a single instrument, Grant Amount$_{it}$, which was equal to the total number of rupees that the microenterprise had received in grants up until time t. They ran the IV regression:

$$\text{Profit}_{it} = \alpha + \beta \cdot \text{Capital}_{it} + \gamma_t + \varepsilon_{it}$$

with the first-stage regression:

$$\text{Capital}_{it} = \theta + \pi \cdot \text{Grant Amount}_{it} + \delta_t + u_{it}$$

They find extremely high returns to capital, averaging 5% per month (about 60% per year).

Exercise 1. What does it mean for the instrument to be relevant? How can we test for relevance?

Exercise 2. Is the grant amount as-good-as-randomly assigned? Why or why not?

Exercise 3. What would the "reduced form" regression be?

Exercise 4. What does the exclusion restriction mean in this setting? Do you think the grant amount is excludable? Why or why not?

Military Service and Earnings (Angrist 1990)

A classic example of instrumental variables is Josh Angrist's 1990 paper on the effect of military service on earnings later in life. Looking at a cohort of men who were eligible for the Vietnam draft, Angrist uses the draft lottery to construct an instrument for military service. For each cohort, the US government randomly drew birthdays to determine who would be draft-eligible: men whose birthday was drawn early would be drafted before men from the same cohort whose birthday was drawn late. Angrist creates a dummy variable equal to one for those whose birthday was draft-eligible, and zero otherwise, and then uses that as

an instrument for military service. He finds, using this IV strategy, that military service lowered eventual earnings by 15%.

Exercise 5. Why do we need an instrument? What might cause military service to be endogenous?

Exercise 6. Do you think that the instrument is as-good-as-randomly assigned? Why or why not?

Exercise 7. What would the reduced form regression be? What is the first stage regression?

Exercise 8. The Vietnam draft had an exemption for college students. Suppose that some young men stayed in college in order to avoid the draft. Would that violate the exclusion restriction? If so, how do you think it might bias the results?

The Oregon Health Insurance Experiment (Finkelstein et al. 2012, Baicker et al. 2013)

In 2008, Oregon began to expand its Medicaid program (Medicaid is a health insurance program for low-income adults; it is administered by state governments, but mainly funded by the federal government). Since they did not have enough funds to cover everyone who was interested, they decided to randomize: out of 90,000 people on the waitlist, they randomly chose to invite 30,000 people to apply. There was imperfect compliance in both directions: some people in the control group ended up receiving Medicaid, and less than half of the treated group ended up enrolling in Medicaid. However, the treatment increased the probability of enrolling in Medicaid by over 25 percentage points.

A team of researchers used this random assignment to study the effects of Medicaid using an IV (Finkelstein et al., 2012; Baicker et al., 2013). They found that Medicaid improved various financial outcomes, and led to improvements in mental health and wellbeing. However, results for effects on physical health were mixed, and Medicaid receipt increased healthcare spending substantially.

Exercise 9. The mean of a dummy variable for ever being on Medicaid was equal to 0.397 for the treated group, and 0.141 for the control group. What is the first-stage coefficient?

Exercise 10. The mean of a dummy variable for experiencing depression symptoms was equal to 0.306 for the treated group, and 0.329 for the control group. What is the reduced-form coefficient?

Exercise 11. Using the results above, what is the IV estimate of the effect of Medicaid receipt on depression?

Exercise 12. Why do we need an instrument? Would you expect an OLS regression of depression symptoms on Medicaid to recover the causal effect of Medicaid?

Instrumental Variables as a Solution for Measurement Error

Instrumental variables can be used to solve the problem of classical measurement error. We showed previously that if X_i is measured with classical measurement error, then we will have attenuation bias: the OLS estimator will be biased towards zero. In general, a valid instrument, which is correlated with the true regressor but not the measurement error, will solve this problem. In fact, if we have two noisy measures of X_i, but the noise in the first measure is uncorrelated with the noise in the second measure, then we can use one measure as the regressor and the other as the instrument, and the IV estimate will be consistent.

Suppose that we are interested in the regression:

$$Y_i = \alpha + \beta \cdot X_i^* + \varepsilon_i$$

but instead of X_i^*, we only observe X_i and Z_i, where

$$X_i = X_i^* + e_i$$
$$Z_i = X_i^* + v_i$$

Assume the measurement error is classical:

$$\text{Cov}\left(X_i^*, e_i\right) = \text{Cov}\left(\varepsilon_i, e_i\right) = 0$$
$$\text{Cov}\left(X_i^*, v_i\right) = \text{Cov}\left(\varepsilon_i, v_i\right) = 0$$

and that the two measurement errors are uncorrelated:

$$\text{Cov}\left(e_i, v_i\right) = 0$$

We then have that OLS is biased towards zero:

$$\begin{aligned}
\hat{\beta}^{OLS} &\xrightarrow{p} \frac{\text{Cov}\left(X_i, Y_i\right)}{\text{Var}\left(X_i\right)} \\
&= \frac{\text{Cov}\left(X_i^* + e_i, \alpha + \beta \cdot X_i^* + \varepsilon_i\right)}{\text{Var}\left(X_i^* + e_i\right)} \\
&= \beta \cdot \frac{\text{Var}\left(X_i^*\right)}{\text{Var}\left(X_i^*\right) + \text{Var}\left(e_i\right)}
\end{aligned}$$

In contrast, using the second noisy measure as an instrument recovers a consistent estimator of β:

$$\begin{aligned}
\hat{\beta}^{IV} &\xrightarrow{p} \frac{\text{Cov}\left(Z_i, Y_i\right)}{\text{Cov}\left(X_i, Z_i\right)} \\
&= \frac{\text{Cov}\left(X_i^* + v_i, \alpha + \beta \cdot X_i^* + \varepsilon_i\right)}{\text{Cov}\left(X_i^* + e_i, X_i^* + v_i\right)} \\
&= \frac{\beta \cdot \text{Var}\left(X_i^*\right)}{\text{Var}\left(X_i^*\right)} \\
&= \beta
\end{aligned}$$

The reason that this works is because our IV assumptions are satisfied. We can rewrite the regression model as follows:

$$\begin{aligned}
Y_i &= \alpha + \beta \cdot X_i^* + \varepsilon_i \\
&= \alpha + \beta \cdot (X_i - e_i) + \varepsilon_i \\
&= \alpha + \beta \cdot X_i + \underbrace{\varepsilon_i - \beta \cdot e_i}_{\text{Error Term}}
\end{aligned}$$

Notice that in this model, the noisily measured X_i takes on a coefficient of β, and the error term contains both the ε_i and the measurement error term. The instrument, Z_i, recovers the true β because it is correlated with X_i but it is uncorrelated with the error term: $\text{Cov}\left(Z_i, \varepsilon_i - \beta \cdot e_i\right) = 0$.

More generally, a valid instrument, which may or may not just be a second measure of X_i, will solve the attenuation bias problem coming from measurement error.

Instrumental Variables as a Solution for Simultaneity

Instrumental variables can be used to solve the problem of simultaneous equations. As we discussed in an earlier lecture, simultaneity and reverse causality will cause a failure of the OLS orthogonality condition, leading OLS to be inconsistent. An instrument can solve the simultaneity problem, if it appears in one of the equations but is excluded from the other.

We will focus on the estimation of the demand curve, which was in fact the original application for which IV was invented. Suppose we have the following system of equations, drawn from different markets, i:

$$\text{Demand: } \log Q_i = \alpha_D + \beta_D \log P_i + \varepsilon_i^D$$
$$\text{Supply: } \log Q_i = \alpha_S + \beta_S \log P_i + \varepsilon_i^S$$

where β_D is the elasticity of demand, ε_i^D is a demand shock, β_S is the elasticity of supply, and ε_i^S is a supply shock. Note that we typically think that β_S is positive and β_D is negative. Rearranging the equations, we have:

$$\log P_i = C + \frac{\varepsilon_i^D - \varepsilon_i^S}{\beta_S - \beta_D}$$

where C is a constant.[1] This shows that in general, log prices are correlated with both supply and demand shocks, and so an OLS regression will recover neither the supply nor demand elasticity.

Suppose that we have an instrument, Z_i, which is correlated with log prices but is not correlated with ε_i^D. By our previous arguments, we know that using this instrument, our IV estimator will identify the elasticity of demand, β_D.

What sort of variable will make a suitable instrument for demand? In general, the key is to isolate a source of supply shocks. In order for the instrument to be relevant (correlated with prices) it must either be a demand shock, a supply shock, or some combination of the two. In order for it to be uncorrelated with ε_i^D, it must be that the instrument is a supply shock, rather than a demand shock. The instrument thus estimates the elasticity of demand by shifting the supply curve in and out, tracing out the demand curve.

[1] We can also derive $\log Q_i = C_2 + \frac{\beta_S \varepsilon_i^D - \beta_D \varepsilon_i^S}{\beta_S - \beta_D}$, where C_2 is a constant.